

■ 1.5.3 Integration

Here is the integral $\int x^n dx$ in *Mathematica*.

```
In[1]:= Integrate[x^n, x]
```

$$\text{Out}[1]= \frac{x^{1+n}}{1+n}$$

Here is a more complicated example.

```
In[2]:= Integrate[1/(x^3 - 1), x]
```

$$\text{Out}[2]= -\left(\frac{\text{ArcTan}\left[\frac{1+2x}{\text{Sqrt}[3]}\right]}{\text{Sqrt}[3]}\right) + \frac{\text{Log}[-1+x]}{3} - \frac{\text{Log}[1+x+x^2]}{6}$$

Mathematica knows how to do many kinds of integrals. It can integrate any rational expression (ratio of polynomials), at least so long as the denominator does not involve too high a power of x . *Mathematica* can also integrate expressions that include exponential, logarithmic and trigonometric functions, so long as the resulting integrals can be given in terms of this same set of functions.

There are however many integrals for which no explicit formulae can be given, at least in terms of standard mathematical functions. Even a seemingly innocuous integral like $\int \frac{\log(1+2x)}{x} dx$ can, for example, only be done in terms of the dilogarithm function `PolyLog[2, x]`. And an integral like $\int \sin(\sin(x)) dx$ simply cannot be done in terms of any of the functions that are defined in standard mathematical handbooks. Section 3.5.7 discusses how *Mathematica* treats different kinds of integrals.

Mathematica can do integrals like this.

```
In[3]:= Integrate[Log[x]^2 (1 + x^2)/x, x]
```

$$\text{Out}[3]= \frac{x^2}{4} - \frac{x^2 \text{Log}[x]}{2} + \frac{x^2 \text{Log}[x]^2}{2} + \frac{\text{Log}[x]^3}{3}$$

This integral simply cannot be done in terms of standard mathematical functions. As a result, *Mathematica* just leaves it undone.

```
In[4]:= Integrate[Sin[Sin[x]], x]
```

```
Out[4]= Integrate[Sin[Sin[x]], x]
```

Here is the definite integral $\int_a^b \log(x) dx$.

```
In[5]:= Integrate[Log[x], {x, a, b}]
```

```
Out[5]= a - b - a Log[a] + b Log[b]
```

Mathematica cannot give you a formula for this definite integral.

```
In[6]:= Integrate[Sin[Sin[x]], {x, 0, 1}]
```

```
Out[6]= Integrate[Sin[Sin[x]], {x, 0, 1}]
```

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You can still get a numerical result, though.

```
In[7]:= N[ % ]
Out[7]= 0.430606
```

<code>Integrate[f, x]</code>	the indefinite integral $\int f \, dx$
<code>Integrate[f, {x, xmin, xmax}]</code>	the definite integral $\int_{xmin}^{xmax} f \, dx$
<code>Integrate[f, {x, xmin, xmax}, {y, ymin, ymax}]</code>	the multiple integral $\int_{xmin}^{xmax} dx \int_{ymin}^{ymax} dy \, f$

Integration.

This evaluates the multiple integral $\int_0^1 dx \int_0^x dy (x^2 + y^2)$. The variables appear in `Integrate` in the same order as in the standard mathematical notation: with the outermost one *first*.

```
In[8]:= Integrate[ x^2 + y^2, {x, 0, 1}, {y, 0, x} ]
Out[8]= 1/3
```